

\$ sih2026 --brief --per-ps

What it asks. What we build. How we show it.

official description → our solution → how we build it →
working flowchart → PPT plan → tips. per problem.

SIH26156 · ULPF / NTRO

SIH26087 · NCCT / Cooperation

SIH26076 · Mausam / IMD

SIH26138 · Green Fleet

deadline 05 oct 2026 · october 2026

0 • WHAT THE WEBSITE ASKS (OFFICIAL, CONDENSED)

Background. Enterprises generate massive log volumes – network devices, servers, OS, apps, databases, cloud, containers, endpoint tools, identity systems, IoT – in Syslog, JSON, XML, CSV, CEF, LEEF, proprietary and app-specific formats. This diversity breaks central monitoring, security ops, compliance, investigation and threat analytics. Teams hand-write parsers before SIEM, lake or ML tools can use anything.

Task. Design a framework that ingests, parses, normalises and standardises logs from **any** hardware or software system – preserving originals for forensics/compliance – into one schema for analytics, correlation, visualisation, threat hunting, anomaly detection and ML. Scalable, extensible, vendor-agnostic, big-data scale (billions of events/day).

It must: (a) keep raw data loss-free • (b) parse source attributes • (c) normalise to a common taxonomy • (d) trace raw→normalised • (e) onboard new sources plug-and-play • (f) unify visibility • (g) feed SIEM + data lake • (h) be AI/ML-ready • (i) cut parser effort • **(j) run air-gapped** • (k) ship as a container.

Scope: any **perimeter device** log → standardised, lossless, analytics-ready form for next-gen SIEM.

Deliverables: code link • README + setup • architecture doc ≤ **2 pages** • demo video ≤ **2 min** • tech presentation ≤ **5 slides**.

source: sih.gov.in/sih2026PS • contact via nciipc.gov.in

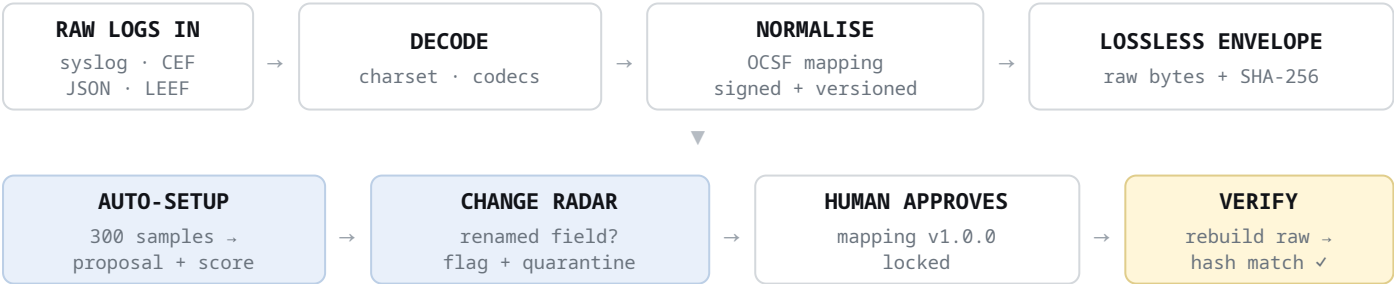
2 • WHAT WE CAN BUILD – LOGSMITH

A pipeline that converts any device log into the open **OCSF** standard and seals the original log inside every output – then **proves nothing was lost** by rebuilding the original from its own output, live. Plus: point it at 300 sample lines and it learns a new device's format by itself (with a score), and it raises an alarm when a vendor update renames a field.

3 • HOW WE BUILD IT (SHORT)

Python pipeline: format readers (syslog / CEF / LEEF / JSON) → template mining that finds each log's structure → a mapping table (versioned file) that converts fields to OCSF → every output sealed with the original bytes + checksum. A small local model only suggests mappings for new fields – a human approves. Control screens in React. Everything runs on CPU inside one container with no internet. Start with 5 common formats, prove it on public log samples.

4 • HOW IT WORKS



5 • PPT – 5 SLIDES (THE PS CAPS IT HERE)

S1	The parser problem	show one real mangled field, dropped by a parser
S2	LOGSMITH, one diagram	the pipeline; the lossless envelope is the hero
S3	Live proof, our numbers	coverage %, verify rate, onboard time - measured
S4	Architecture + air gap	one container, CPU-only, zero network calls
S5	Deploy path + ask	what works, what's next, the samples we need

6 • TIPS + OUT-OF-THE-BOX IDEAS

- **Verify, don't claim.** Rebuild-and-hash live is the whole pitch - rehearse till it cannot fail.
- **Rename a field mid-demo.** Let the radar catch it. Motion beats slides.
- **Never ship OCSF 2001.** Deprecated - current classes prove you read the standard.
- **Air-gap proof.** Network panel open, zero calls. End at the stream; never build a SIEM.

02 SIH26076 – MAUSAM HOMEPAGE • IMD

Personalised homepage for the 'Mausam' mobile application • Software • Smart Automation • 178 slots left – most crowded

0 • WHAT THE WEBSITE ASKS (OFFICIAL – THIS PERSONA LIST IS THE WHOLE STATEMENT)

- **Health-conscious:** AQI, pollen, UV, humidity – allergies, asthma, skin.
- **Fitness:** sunrise/sunset, “best running hours”, wind, heat alerts.
- **Beach & surf:** sea state, tides, wave height, water temperature.
- **Travellers:** saved places, severe alerts for flights, packing hints (“carry a raincoat in London”).
- **Parents & families:** school commute, rain alerts, severe warnings.
- **Agriculture:** soil moisture, rain outlook, frost alerts, planting guidance.
- **Commuters:** weather + traffic, visibility, storm/fog alerts.
- **Event planners:** extended forecast, rain probability, “comfort index”.

source: sih.gov.in/sih2026PS • that is the complete official text – no scope, dataset or API list beyond it

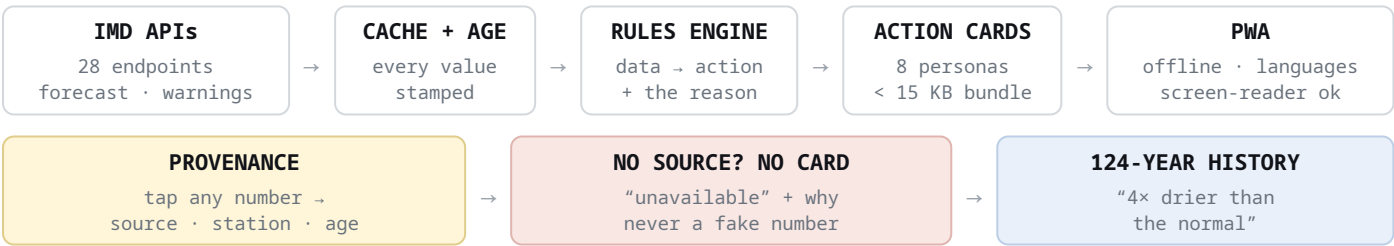
0 • WHAT WE CAN BUILD – MAUSAM (THE PERSONALISED HOMEPAGE ITSELF)

The Mausam home screen, rebuilt as a decision layer: each user gets **actions**, not charts (“heat-wave warning + 44°C → stay indoors 12-4”). Every number tap-reveals its exact IMD source, station, issue time and age, and every advisory is deterministic enough to be re-derived by hand. Works offline on slow networks, in regional languages, and honestly says “data unavailable” where IMD publishes nothing (pollen) instead of inventing a number.

0 • HOW WE BUILD IT (SHORT)

A small Python service on a **static-IP host** holds the IMD API key – the key is IP-bound and every call needs a user JWT that expires hourly – and polls the 28 documented endpoints, with the public warning feed as backup. A rules engine turns fresh data into action cards with the reason attached. The home screen is an offline-first web app that caches your personal forecast on the phone, works on 2G, and reads warnings aloud for accessibility. Maps use the free Leaflet library.

0 • HOW IT WORKS



0 • PPT – THE OFFICIAL 6-SLIDE IDEA FORMAT (SIH TEMPLATE, TITLE INCLUDED)

S1	Title	team • SIH26076 • exact title • MoES/IMD • Software • Smart Automation
S2	Problem	eight lives, one app; IMD publishes data, not decisions
S3	Proposed solution	the decision home screen; provenance toggle is the hook
S4	Technical approach + feasibility	28 IMD endpoints; key + hourly JWT on a static-IP host; replayable rules
S5	Impact	agri • commuter • health deep; honest gap cards; no invented stats
S6	Research + references	IMD API guide, CAP/SACHET, CPCB, 124-yr climatology

0 • TIPS + OUT-OF-THE-BOX IDEAS

- **The toggle is the demo.** One button exposing every number's source and age.
- **Throttle to 2G on stage.** Loading from cache while rivals spin wins.
- **Show a gap card proudly.** Pollen: not published by IMD. Restraint reads as skill.
- **Never pitch lightning alerts as new.** They already ship in Mausam - IMD judges know.

03 SIH26087 – COOPERATIVE ECOSYSTEM • NCCT

AI & LMS-enabled capacity building, ERP & employment ecosystem • listed as Hardware, statement says Software + Hardware • 402 slots left – least crowded

0 • WHAT THE WEBSITE ASKS (OFFICIAL, CONDENSED)

Background. NCCT (VAMNICOM + 5 RICMs + 14 ICMs) trains cooperative staff, PACS, SHGs, dairy co-ops, farmers, rural youth. Trained youth still can't reach jobs or visible certification. Systems are manual: no tracking, no analytics, weak job linkage.

Task. One central web platform combining ERP training management + e-learning + analytics + skill development + digital literacy + employment exchange for cooperative stakeholders and rural youth.

Features demanded: online registration • trainee profiles • multilingual e-learning • attendance (face/QR) • timetable/hostel/logistics • LMS + assessments + certification • certificate verification • career chatbot • employer dashboard • mobile + offline learning • central outreach database.

Tech: LMS • cloud ERP • analytics • mobile apps • interactive media • face/QR. **Mode:** Software + Hardware.

source: sih.gov.in/sih2026PS • scale reference: NCCT annual report 2023-24 – 3,759 programmes, 2,27,177 trained

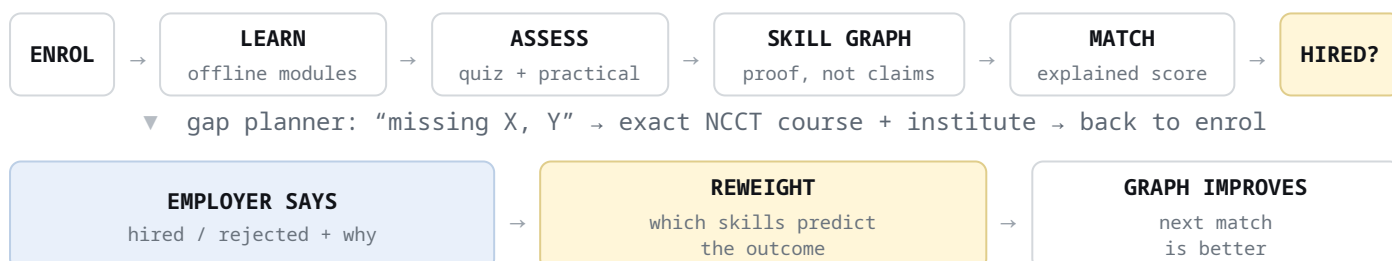
0 • WHAT WE CAN BUILD – SAHAKAR SETU

Skip eleven shallow tabs. Own one gap: **course** → **proven skill** → **job**. Skills need evidence (quizzes, practicals, attendance) – no evidence, no claim. Jobs match on proof, each gap linked to the exact NCCT course closing it. Employers log hired/rejected + why; the system learns what predicts outcomes. Certificates verify on a public page. Attendance QR-first; face optional, consented, on-device.

0 • HOW WE BUILD IT (SHORT)

Offline-first web app (no internet needed, syncs later, Indian languages) + Python back-end with trainees, courses, scores, jobs. Quizzes feed a scoring engine (skill + confidence range). Matching is rules-based and explainable. QR attendance on ordinary phones. Seed with 3 syllabi.

0 • HOW IT WORKS



0 • PPT – SLIDE BY SLIDE

S1	Scale, zero proof	20 institutes, 2.27L trained, no verified skill
S2	Why portals fail	keyword-match demo with nonsense scores
S3	Evidence pipeline	the flow above, one diagram
S4	Graph + attestation	skill node: evidence, confidence, fingerprint
S5	Explainable matching	one match, every point accounted for
S6	The outcome loop	hire/reject reasons reweight the graph live
S7	Offline + honesty	QR-first, consent-led biometrics, shared devices
S8	Architecture	app + API + DB, sync, public verify page
S9	Pilot + rollout	match-to-interview, gap-closure, 20 institutes
S10	Demo + ask	the 60-second loop; syllabus + SPOC ask

0 • TIPS + OUT-OF-THE-BOX IDEAS

- **The 60-second loop is the demo.** Quiz to skill to match to reject to reweight.
- **Lead with privacy.** QR-first, face optional with consent - before anyone asks.
- **Let a judge verify a certificate.** On their own phone. Checkable beats claimable.
- **Map 3 programmes deep.** Not 30 shallow. Borrow their motto: Sahakar se Samridhi.

0 • WHAT THE WEBSITE ASKS (OFFICIAL, CONDENSED)

Background. Maritime and logistics face pressure to cut greenhouse gases while staying efficient and cheap. Fuel is among the largest costs and impacts. Traditional methods struggle with its high-dimensional, non-linear nature. Quantum-inspired metaheuristics (quantum search ideas, classical computers) are the proposed way forward.

Task. Predict fuel use per vessel type and condition; optimise vessel mix, capacity, speed; fold in LNG, methanol, hydrogen, ammonia + shore power; minimise fuel, cost, lifecycle emissions under demand, reliability, emission rules.

Objectives: quantum-inspired fuel prediction • quantum-metaheuristic fleet optimisation • minimise fuel + cost + lifecycle GHG • meet demand, reliability, regulations • **benchmark against conventional methods** (accuracy, convergence, quality, scale).

Expected: a software platform – modelling, data-driven prediction, multi-objective optimisation, constraints, alternative-fuel scenarios, benchmarking with case studies.

source: sih.gov.in/sih2026PS • context: Egreen Quanta × Mormugao Port quantum-maritime agreement, Jul 2026

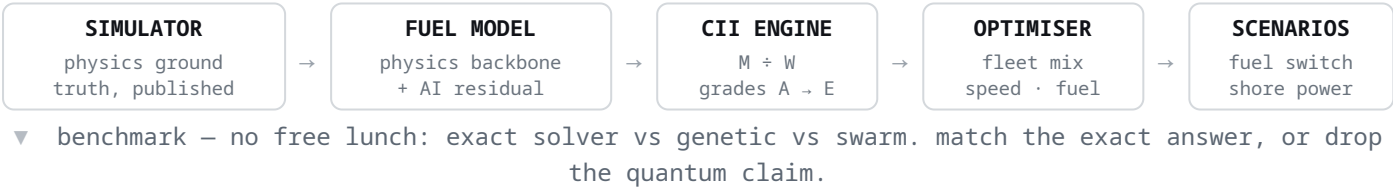
0 • WHAT WE CAN BUILD – SEAWARD

Four moves: **physics first** (small, explainable, works on unseen ships), AI only on leftovers; **CII-native optimisation** (optimise the actual IMO A-E ship grade, not just fuel); **honest “quantum-inspired”** (classical code, checked against an exact solver on small fleets – no match, no claim); and a **cliff slider** showing today's best fleet illegal by 2030, re-optimised with its price.

0 • HOW WE BUILD IT (SHORT)

Python (NumPy/SciPy): ship physics (power vs speed, wind/wave/fouling), a few fitted params per ship type, gradient boosting on residuals only. A voyage simulator is the test ground – real ship data is anonymised by IMO, so the simulator is the honest dataset. Quantum-inspired search for mix × speed × fuel, cross-checked by exact integer programming. Front-end: map, charts, cliff slider. CPU-only, fixed seeds.

0 • HOW IT WORKS



0 • PPT – SLIDE BY SLIDE

S1	The maritime trilemma	fuel cost vs emissions vs compliance
S2	Black boxes fail	confidently wrong outside their training range
S3	Physics + residuals	the model in one diagram; who owns what
S4	CII-native	the real IMO formula; grade, not tonne
S5	Honest quantum	named mechanisms, exact-solver check on screen
S6	The cliff deck	2030 slider: collapse, re-optimize, capex
S7	Measured benchmarks	same harness, same seeds, our table
S8	Architecture	simulator as dataset, equations published
S9	Pilot path	metrics, Mormugao data ask, 2027 readiness
S10	Demo + ask	the slider moment; what we need next

0 • TIPS + OUT-OF-THE-BOX IDEAS

- **The slider is the demo.** 2030, collapse, re-optimize. One motion, whole story.
- **Ablation on screen.** Physics vs +AI vs black-box proves the architecture.
- **Say the data truth out loud.** Ship data is anonymised by design; here is our simulator.
- **Rs/tonne-CO2-avoided on every scenario.** The number ports decide on.